

ORIGINAL RESEARCH

Road damage detection with bounding box and generative adversarial networks based augmentation methods

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Abstract

In this paper, based on the data augmentation techniques of bounding box augmentation and the road damage generative adversarial network based augmentation, a robust road damage detection method has been presented. To this end, first, Iran road damage dataset has been collected by means of a dashboard-installed mobile phone. After processing these images by the blind referenceless image spatial quality evaluator technique, the substandard and inferior data have been automatically eliminated. In the second step, based on the YOLOv5 with several different baseline models, an algorithm has been developed for detecting the road surface damages. In the third step, by using the traditional as well as the bounding box augmentation and road damage generative adversarial network based augmentation techniques, the precision and the robustness of road damage detector under different environmental and field conditions have been improved. Finally, through the ensemble of the best models, the final detector accuracy has been enhanced. The findings of this article indicate that by using the proposed approach, the values of mAP and F1-score are improved by 13.79% and 7.58%, respectively. The dataset and parts of the code are available at: <https://github.com/IranRoadDamageDataset/IRRDD>.

1 | INTRODUCTION

The surfaces of roads and motorways gradually deteriorate because of the damages caused by different factors such as harsh climatic conditions, large volumes of traffic, the age of a roadway, and the type of material used in its construction [1]. These damages increase the risk of driving accidents and the cost of vehicle maintenance [2]. The proper maintenance of roads, which requires correct periodic inspections of such infrastructures, plays an important role in the economic progress of a country [3]. Moreover, knowing the conditions of roadway surfaces can help the navigation system in autonomous vehicles to choose more suitable routes and driving speeds for greater passenger comfort [4]. Services such as Google Maps and Waze can use this information to suggest alternative routes of higher quality to drivers [5]. In most countries, human labour is involved in checking the road surfaces and determining their condition [6]. Such road inspections are carried out either on foot or in vehicles moving at low speeds [7]. This process is very time-

consuming and depends largely on the opinions and discretions of the road inspectors. For these reasons, the automatic road inspection techniques, which enjoy greater speed and detection accuracy and lower cost, have attracted the attention of many entities involved in road inspection and maintenance works. The detection algorithms that are based on vibration sensors (e.g. gyro) can detect road damages such as potholes by measuring the amplitude variations. These sensors have a low precision and are affected by factors such as vehicle speed and suspension system [8]. The laser-based sensors have a high cost and cover only a limited portion of a road [9, 10].

The recent advances in the fields of artificial intelligence and deep learning have led to an extensive use of these methods in the detection and classification of objects [11]. These models, which are based on convolutional neural networks (CNNs), have been recognized as the most efficient road damage detection approaches [12]. CNNs are employed in the tasks such as the detection of the driving and street signs, street cleaning, detection of the driving lines and pavement markings,

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and they have performed much better than the other methods. For the first time, Maeda et al. [11] developed a software program for road damage detection in Japan. In 2018, they used a cellphone camera to compile the first road damage detection dataset. Using this dataset, the first road damage detection challenge was undertaken in the same year with the participation of the big data conference. Experimental investigations showed that the models developed based on this dataset do not have a comprehensive performance under realistic conditions [13]. To solve these problems, the RDD2020 dataset [14] was subsequently presented. In the big data conference of the year 2020, RDD2020 dataset was used as reference for the global road damage detection challenge (GRDDC) [13].

The existing road damage detectors have a low accuracy [7]. The RDD2020 dataset has a small number of annotated images and it does not cover different environmental conditions. To develop a comprehensive and effective road damage detection model, we would need more annotated images than those offered by the RDD2020 dataset. These data should include various environmental conditions such as different shades and illumination levels, sun flares etc. Collecting pictures of road damages by means of a dashboard-installed mobile phone is an inexpensive way of detecting the current road damages. But due to the vibrations and the existing physical state of the vehicle used in the process, this method generates a lot of inferior and low-quality data. The elimination of these data via the manual examination of each image produced in this way is a costly and lengthy process. If cellphone cameras are to be employed in the detection of road damages, an automatic approach should be devised for omitting the inferior images produced by these cameras.

The presented approaches in GRDDC have mainly focused on enhancing the precision of road damage detectors under normal and ordinary conditions. These detectors have not been tested under different environmental and field conditions, and such factors could reduce their accuracy under realistic circumstances. For example to analyze the effect of shadows on road images, various lighting and brightness conditions at different times of the day should be created. Images with such features should be taken within limited time duration. These images should then be labelled manually and very accurately. In order to have a robust database, we need to compile its constituent data under all the different operational conditions a damage detector may face. Training a model by means of a robust dataset makes that model robust against environmental variations. Producing a naturally robust database is a very time-consuming and expensive undertaking. Recent research works on autonomous vehicles and influential road factors have focused on improving the performance of object detectors in difficult and challenging environmental conditions [15–17]. One of the main approaches for creating a robust dataset is to use data augmentation techniques. These techniques enhance the generalizability of damage detectors in various ambient and environmental conditions. Moreover, in the case of limited data, the data augmentation methods prevent the overfitting of the detector models. In this paper new data was generated using traditional data augmen-

tation and combining generative adversarial network (GAN) [18] based data augmentation and bounding box augmentation (BBA) with modified Poisson blending [19]. We have used traditional data augmentation techniques in order to generate robust training data. These data include environmental conditions such as the shadows cast on road surface by different objects, intense sunshine, and brightness variations. Maeda et al. [20] report that they had to generate the synthetic training data manually via image blending operations. This method is very time-consuming and makes it difficult to create GAN based synthetic datasets. In this paper, by presenting an automated blending algorithm based on the modified Poisson blending, the synthetic data of GAN models and BBA transformations are automatically combined with original images. In addition, by using StyleGAN2-ADA, it is possible to create artificial road damages with a limited number of training samples for the GAN model [21].

Also, in the published papers, no method has been implemented for improving the process of data collection and eliminating the low-quality data. An appropriate approach in this respect could generate high-quality data and increase the number of road damage samples. Therefore, a new dataset has been developed in this paper for this purpose. The blind referenceless image spatial quality evaluator (BRISQUE) [22] method is used to get rid of the low-quality and substandard data in this dataset. By examining some state-of-the-art object detectors, the YOLOv5 [23] is employed to create the baseline road damage detection models. Overall, by presenting a new dataset and a data cleaning pipeline, we intend to deal with the existing challenges in the area of road damage data collection via mobile phones. Also, by introducing several models that have been trained by different data augmentation approaches and by evaluating their performances, we attempt to present an effective road damage detection method.

The main achievements of this research can be summarized as follows:

1. Collecting a local dataset of road damages in Iran, which includes various ambient conditions such as different shades, illumination levels, and daylight hours.
2. Eliminating the low-quality images from the annotation cycle via BRISQUE approach.
3. Introducing a novel data augmentation technique using BBA and modified Poisson blending.
4. Generating a new training data using GANs and modified Poisson blending.

The rest of this paper is organized as follows. A review of the papers published on the subject of road damage detection is presented in Section 2. The proposed dataset is introduced in Section 3. In Section 4, first, the presented baseline models are described and then the introduced data augmentation methods are outlined. The proposed method is compared with the other approaches in Section 5. Finally, the conclusion is presented in Section 6. Table 1 summarizes the list of abbreviations.

TABLE 1 List of abbreviations.

Abbreviation	Full writing
ARDB	Automatic road damage blender
BBA	Bounding box augmentation
BRISQUE	Blind referenceless image spatial quality evaluator
CLAHE	Contrast limited adaptive histogram equalization
CNNs	Convolutional neural networks
FLOPS	Floating point operations per second
GAN	Generative adversarial network
GRDDC	Global road damage detection challenge
IoU	Intersection over union
IQA	Image quality assessment
IRRDD	Iran road damage dataset
KID	Kernel inception distance
NMS	Non-maximum suppression
RDGBA	Road damage generative adversarial network based augmentation
SSIM	Structural similarity index measure
SSD	Single shot multi-box detector
SVM	Support vector machine
SVR	Support vector regression
TTA	Test time augmentation

2 | LITERATURE REVIEW

Many of the road damage detection models use a combination of the machine learning methods and image processing techniques for damage detection purposes. Significant progress has been achieved in recent years by merging these two approaches. The image processing-based techniques have achieved significant advances in their feature extraction capabilities [11]. In these models, the 2D images are processed for the extraction of features such as cracks. In image processing approaches, cracks are detected and segmented by means of the thresholding technique and the multi-scale wavelet transform approach [24–26]. Zalama et al. [27] have used a Gabor filter for the detection of surface cracks. Chun et al. [28] employed the image processing techniques and a Bayes classifier to classify various road damages. In the more recent methods, the image processing techniques are combined with machine learning in order to achieve a higher precision. For example by combining two local binary patterns with a support vector machine (SVM) classifier, Chen et al. [29] enhanced the precision of road crack detection. Under the effects of factors such as the illumination and the non-uniform texture of asphalt, these methods produce erroneous results [30–32].

The deep learning methods have resulted in the highest object detection and classification accuracies [33]. The object detectors are grouped as the region-based convolutional neural networks (R-CNNs) and the YOLO detectors [34, 35]. The R-CNN and the Faster-RCNN [36] models have a heavy processing load. These models are not suitable for real-time

applications and cannot be used in mobile systems. The single shot multi-box detector (SSD) [37] enjoys a very high detection speed. The SSD has been employed in conjunction with the MobileNet [38] for the detection and classification of road damages [7]. The Faster-RCNN has been used in ref. [39] for the detection and classification of various types of structural damages. The SSD-based detectors in this research have a very low accuracy. The models which are based on the R-CNN, Faster-RCNN, and R-FCN have improved the accuracy of the structural damage detectors [40, 41]. Many deep learning-based methods are used for the detection of road damages.

Llopis-Castelló et al. [42] determined the type and severity of road damages after detecting the damage zones. In 2020, the researchers employed the U-Net architecture and the YOLOv3 algorithm to detect and classify various types of road damages in China [43]. Using a new backbone, Wan et al. [44] presented the YOLO-LRDD algorithm. They increased the quantity of data in the RDD2020 dataset by using the data collected in China. The method proposed by these researchers boosted the detection speed by 22%. The existing problems with road damage datasets such as the inconsistent resolution and the low-quality of their data have also been highlighted. Han et al. [45] showed that their proposed approach (i.e. CrackW-Net) achieves the best crack segmentation performance in comparison with the U-Net, FCN, and the ResU-Net networks. The authors in 2019, employed a pixel-level classification method for detecting the road surface cracks. They also improved the classifier performance by assigning weight to cost function and using data augmentation techniques [46].

Yanbo et al. [47] used the Faster-RCNN and the SSD to detect the road damages. By employing the Faster-RCNN and tuning the hyper-parameters such as the anchor box ratio and performing various data augmentation techniques (e.g. variation of brightness, Gaussian blur), Yanbo et al. [47] were able to enhance the accuracy of their detection model. The YOLOv3 [48] was used along with the DarkNet53 by Alfarrarjeh et al. [49] to detect the road damages. These authors also attempted to increase the detection accuracy of the models by augmenting the data for the classes with insufficient images. Also, Kluger et al. [50] employed the Faster-RCNN, the RetinaNet [51], and the random forest algorithm to deal with the data scarcity problem. Maeda et al. [20] applied the GANs [18] to create synthetic potholes. First, they used the PG-GAN [52] to produce the potholes and then employed the Poisson blending [53] technique to blend these potholes into the images that showed no damage. This enhanced the accuracy of their road damage detector in the pothole class. The point to consider in the Poisson blending approach is the creation of masks for the blending operations. This is a time-consuming process and it makes the compilation of a large artificial dataset a difficult task. Cao et al. [54] evaluated different SSD and R-CNN models and discovered that the SSD inception V2 and the Faster-RCNN models have the best performance in terms of speed and accuracy.

In 2020, 121 teams participated in the GRDDC 2020 [13]. The goal of this challenge was to present different road damage detection methods based on deep learning. The winner of this challenge was the IMSC team [10], which achieved the best

accuracy and obtained the highest score by using the YOLOv5 and the “ensemble prediction”, “ensemble learning” and the “test time augmentation (TTA)” models. The methods that use the ensemble models enjoy a greater accuracy, but are burdened by a large processing load. In GRDDC 2020, the Cascade R-CNN algorithm has been used for road damage detection by applying the techniques of ‘model ensembling’ and ‘consistency filtering mechanism’ [55]. The approaches presented in the GRDDC 2020 have used models with very high numbers of parameters. The superior algorithms in this challenge have used the ‘ensemble modelling’ approach to enhance the final detection precision. These methods cannot be implemented in systems of low processing power. The RDD2020 dataset was examined, and it was shown that the merging of data from all countries does not boost the accuracy of the detectors; it only enhances the generalization of the detection models [7].

The MN-YOLOv5 algorithm was presented by Gue et al. [56] and they were able to reduce the processing load and the size of the baseline model used for the detection of road damages. In 2022, image information and inertial sensor data from a mobile phone were combined for road damage detection [57]. The processing of such information enables the detection of pothole in different weather conditions by means of machine learning techniques and CNNs. The main problem of the deep learning-based approaches used for road damage detection is the shortage of relevant data in different classes. The StyleGAN model and the Road-Seg-CapsNet method were employed in ref. [58] to deal with the problem of edge information loss. Their proposed approach improved the segmentation accuracy of road cracks. Using a real-time approach, Choi & Cha [59] presented the SDDNet model for the segmentation of concrete cracks. This method achieved higher accuracy in the evaluation metrics by reducing the number of model parameters. To enhance the performance of road damage detectors, the ‘ghost attention’ and the ‘proposal feature metric’ modules were employed in ref. [60]. The proposed method improved the detection accuracy of rare road damages. Silva et al. [61] conducted road damage detection operations by using different versions of the YOLO algorithm and the road damage images collected by a UAV. The attention-based generative adversarial network has been used in ref. [62] to deal with the challenge of creating a dataset. In similar research, Guan et al. [63] used stereo vision along with segmentation analysis to detect pavement cracks and potholes. Some research works have used sensor information instead of the image processing techniques in order to detect the road damages. For example accelerometers and mobile phone sensor data have been utilized to check the road conditions [64, 65]. Finally, a summary of the recent approaches to road damage detection is presented in Table 2.

3 | THE PROPOSED DATASET

In this section, first, the existing datasets in this field are surveyed. Next, our dataset is described. Then, the procedure of collecting this dataset, its annotation process, and the manner of evaluating the image qualities are explained.

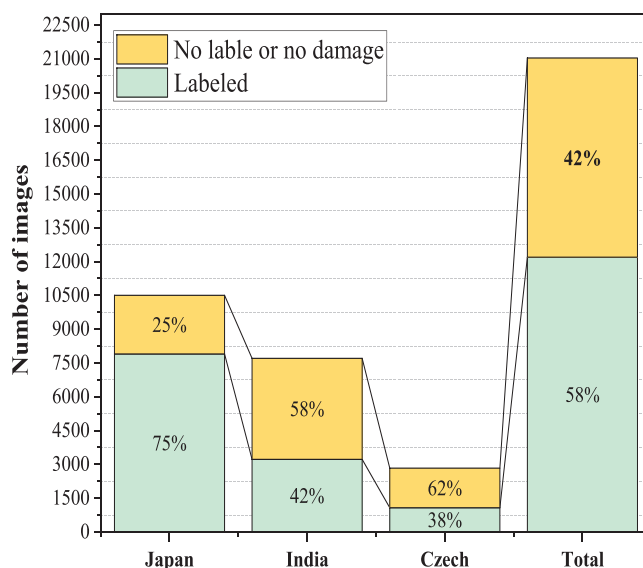


FIGURE 1 Distribution of annotated images in the RDD2020 dataset.

3.1 | A survey of the road damage detection datasets

In 2018, a method and dataset for road damage detection was published [11]. The researchers divided the road damages into eight classes. This dataset contained 9053 images showing 15435 annotated damages. Presented models, had a low recall in most of the classes [11]. This dataset includes unbalanced classes. Some classes, comprised a very small number of images; so by adding extra images, researchers tried to make this dataset more balanced. For example, Angulo et al. [68] increased the size of this dataset to 18,034 images. Their research revealed that the precision and the recall of the detectors are boosted by increasing the number of images and using the data augmentation techniques. In 2019, Maeda et al. [20] upgraded the RDD2018 dataset and presented the RDD2019 dataset, in which the number of annotations and number of images were changed to 30989 and 13135, respectively.

In 2020, the RDD2020 [14] was created based on the RDD2019. This dataset contains 26620 road damage images from the countries of Japan, India and the Czech Republic. Contrary to the former datasets, this dataset comprises four classes. Many environmental and ambient conditions such as shadows, intense sunlight reflections and different lighting conditions are present in this dataset in a very limited number. Also, many images in this dataset are not annotated and show no sign of road damage. The distribution of these images is illustrated in Figure 1. 42% of the images in this dataset show no damages or have no annotations. Moreover, the class distributions in the countries of India, Japan, and the Czech Republic are not equal, and this makes it difficult to establish a transferable model for all the countries. The creation of robust road damage detection models requires a lot more data, which should reflect a great diversity of scenery and ambient conditions. Evaluating

TABLE 2 A summary of the different road damage detection methods classified based on the category of detection.

Reference	Year	Algorithm	Summary of method	Category of detection
Zhue et al. [57]	2022	SVM	- Classifying the images into rainy and non-rainy scenes - Using the inertial sensor data and image information to detect manholes in different environmental conditions	Potholes
Anandhalli et al. [66]	2022	YOLOv3	- Collecting pothole images in different traffic situations - Evaluating the performances of deep learning models in terms of processing power	Potholes
Kang et al. [67]	2022	STRNet	- Real-time crack segmentation by using a semantic transformer representation network (STRNet) - Considering complex scenes for crack segmentation - Using mIoU as an evaluation metric	Cracks
Fang et al. [46]	2021	U-Net	- Dealing with the non-uniform distribution of data by using the weighted cross entropy loss and data augmentation for pixel-level classification	Cracks
Liu et al. [43]	2020	YOLOv3, U-Net	- Crack detection by means of the YOLOv3 algorithm - Crack segmentation by using the U-Net with the pre-trained ResNet-34 network	Cracks
Wan et al. [44]	2022	YOLO-LRDD	- Enhancing the detection speed by means of the Shuffle-ECANet - Improving the understanding of features in a network by means of the BiFPN - Using the Focal-EIOU as a localization loss cost function	Road damages
Guo et al. [56]	2022	MN-YOLOv5	- Using the MobileNetV3 backbone for reducing the number of model parameters - Using the attention module for improving the detector precision	Road damages
Maeda et al. [20]	2021	SSD MobileNet, SSD Resnet50	- Generating synthetic data by means of PG-GAN and Poisson blending - Expanding and the reannotation of the RDD2018 dataset	Road damages
Hegde et al. [10]	2020	YOLOv5	- Combining the two approaches of ensemble prediction and ensemble model along with the TTA for improving the detection accuracy	Road damages
Ours	—	YOLOv5	- Developing a general dataset by using the automated IQA process - Developing an automated procedure for creating a synthetic road damage dataset based on GANs - Presenting a novel approach for BBA - Evaluating the real-time road damage detection models by means of various data augmentation techniques	Road damages

the quality of the collected data is one of the main challenges in this process.

The findings of the former research works and the GRDD2020 challenge indicate that the RDD2020 dataset is not sufficient for developing a comprehensive road damage detection model. The RDD2020 dataset contains images with wrong labels. These incorrect labels diminish the accuracy of the damage detectors. Using this dataset for detecting the road damages in Iran will not yield appropriate results, due to the different camera angles, different field settings of road damages, and the different road paths and driving scenes. Also, the vibration sensors can only detect one type of damage. These issues show that more research is needed in order to achieve a suitable model with adequate accuracy and real-time speed. Moreover, the limited size of the presented datasets makes it difficult to come up with an exact and effective method. For this reason, every country needs a unique local dataset for improving the accuracy of road damage detections in that country. These data

should include sufficient number of annotations in each damage class.

3.2 | Iran road damage dataset

The Iran road damage dataset (IRRDD) includes many pictures of road damages from the cities of Tehran and Karaj in Iran. The routes considered in collecting these images contained the highest numbers of road damages. These images were recorded at different times of the day and under diverse weather and climatic conditions. The selected routes include the local streets, highways and city pathways. The presented dataset has been produced based on the data annotation procedures in the RDD2020 dataset. The only difference is the manner of annotating this dataset. In this dataset, the YOLO format has been adopted for data annotation. The IRRDD is collected via the camera of a dashboard-installed mobile phone. The collected

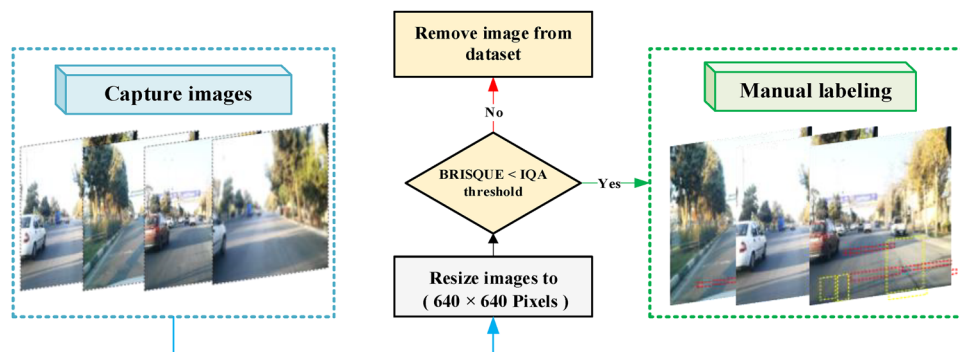


FIGURE 2 Image qualities and the labelling pipeline.

TABLE 3 Distribution of the instances quantities in the IRRDD classes.

Number of instances	Class name	Damage type
25821	D00	Longitudinal crack
8201	D10	Lateral crack
10257	D20	Alligator crack
2776	D40	Pothole

images include the annotations related to the D00, D10, D20, and D40 road damage classes, which have been listed in Table 3. The average car speed during the data collection operation was 40 km/h.

Image quality is a concept which is influenced by the opinions of observers. The image quality assessment (IQA) measure tries to quantify an observer's perception of image quality. This subject is divided into two main research domains. In the first, high-quality reference images are used to measure this criterion. An example of this approach is the structural similarity index measure (SSIM) [69]. In the second approach, there is no need for a reference image to assess the quality of an input image. In these methods, the features that describe the image structures are computed. Then by using the relationship between these features and human opinion, the IQA quantity is determined. Many of the methods in this field use transformation techniques such as the discrete cosine transform and wavelet transform to compute the image features. In the BRISQUE approach, image pixels are used to calculate the image features. In this method, there is no need to perform image transformations.

The images of this dataset are collected at the 640×640 resolution. All these images are annotated under a pipeline. The goal of this pipeline is to eliminate all the low-quality images from the annotation cycle. This method automatically discards the images that have a lot of noise and blur from the annotation process. This improves the quality of the dataset and reduces the workloads of manual annotation and IQA. The diagram of this pipeline is depicted in Figure 2.

In this pipeline, first, all the images collected via cellphone camera are reduced in size with respect to the size of the images in the dataset. Then, the BRISQUE method is employed to

assess the image qualities in terms of blur, noise etc. Based on a support vector regression (SVR) model that was trained on an image dataset with corresponding differential mean opinion score, this algorithm produces BRISQUE scores. This dataset should include images with specific noises (e.g. blurring, compression artefacts) as well as the images without such noises. The images scored in this way must have one of the considered distortions by which the model has been trained. This method assigns a score to an image. The higher this score, the lower the image quality. In the IQA process, first, 100 images were manually selected as the high-quality images. Then the BRISQUE scores were calculated for these images. The highest BRISQUE score for these 100 images was 28. Next, 100 low-quality images were selected and their BRISQUE scores were determined. The average BRISQUE score for these 100 images was higher than 28. Thus, the number 28 was considered as the IQA threshold between the high-quality and low-quality images. All the images whose scores were greater than 28 were eliminated from the datasets. Figure 3 shows some of the images that have been removed by this method. Following the automatic IQA process, the images are annotated manually. In the next step, these images and their annotations are reviewed and analyzed by the road damage detection experts. Finally, these images are ready for training the road damage detection models. Some sample images from the IRRDD have been displayed in Figure 4.

Compared to the other road damage detection datasets, the IRRDD presents a greater number of images from each class. Also in this dataset, road damages are covered by 70% of the collected images. In contrast to the RDD2020, the IRRDD has used a fixed data collection technique. For example in the RDD2020, in the pothole class, the camera angle of some images differs greatly with the camera angle of most of the dataset images.

4 | OUR PROPOSED METHOD

In this section, we describe the baseline object detection models developed for the detection of road damages in the IRRDD. We then present the data augmentation methods for enhancing the precision of these detector models. The block diagram

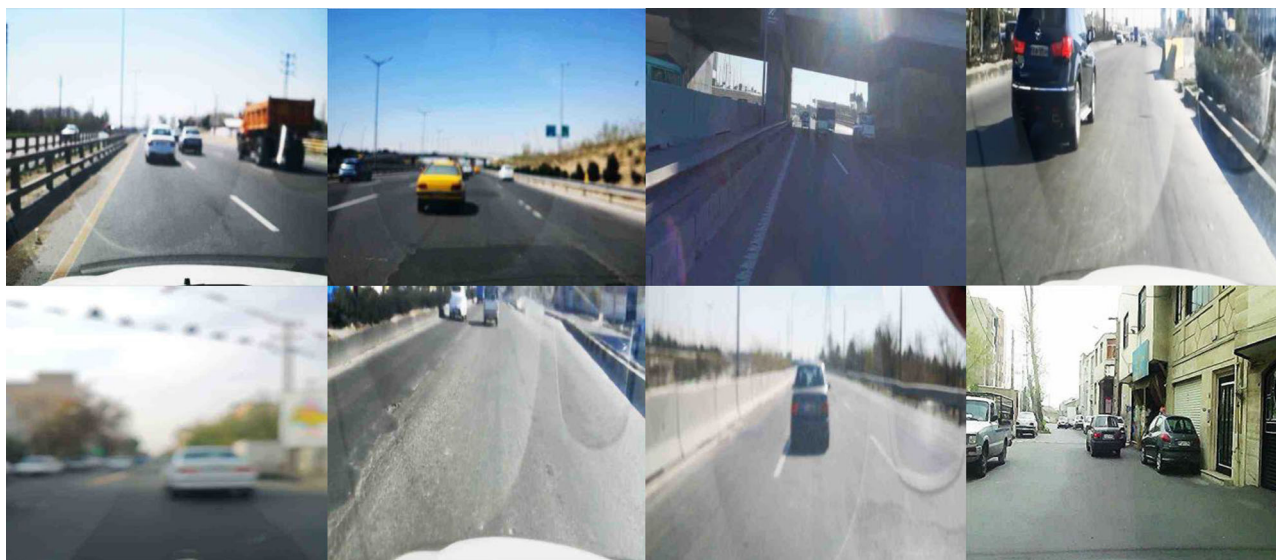


FIGURE 3 Samples of the removed images based on the IQA threshold.

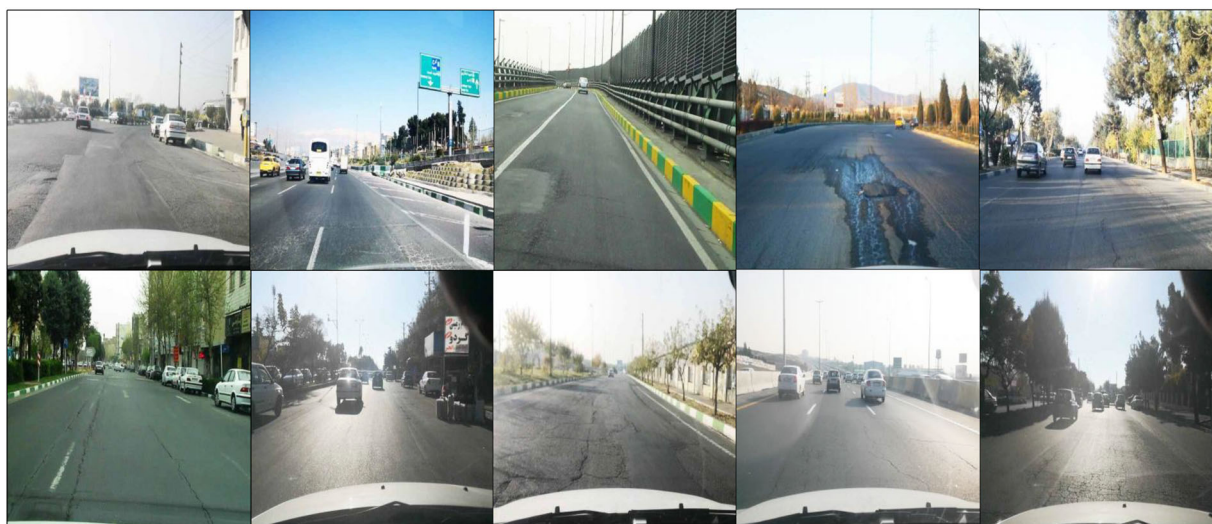


FIGURE 4 Sample images from the IRRDD.

in Figure 5 shows that in the first step, road damage images are collected, their quality is assessed, and the low-quality samples are eliminated. Then these images are labelled and a dataset is established for this paper. In the second step, by employing the approaches based on GANs, BBA, and traditional augmentation techniques, a synthetic dataset is created. In the third step, by aggregating the original and the synthetic datasets, the training data needed for evaluating the performance of the proposed methods are prepared. In this step, first, the baseline models are trained with the original dataset and used for comparing the effects of different data augmentation techniques. Then each model corresponding to a particular augmentation technique is trained. In the final step, these models are evaluated with/without the consideration of TTA.

4.1 | Baseline object detection models

From the standpoint of real-time performance, the one-stage detectors are more optimized than the two-stage ones. Numerous works have employed different types of one-stage models for the detection of road damages. YOLO is the most famous one-stage object detection algorithm. YOLO has evolved in the course of time and different versions of this algorithm (e.g. YOLOv3 [51], YOLOv4 [70], and YOLOR [71]) have been presented. The YOLOv5 uses a combination of the CSP-DarkNet53 and the focus layers. The focus layers reduce the processing load of the YOLOv5 relative to the other models [72]. Three models were initially examined in terms of their speed and accuracy to select a baseline model. The evaluation results of the baseline models have been reported in Table 4.

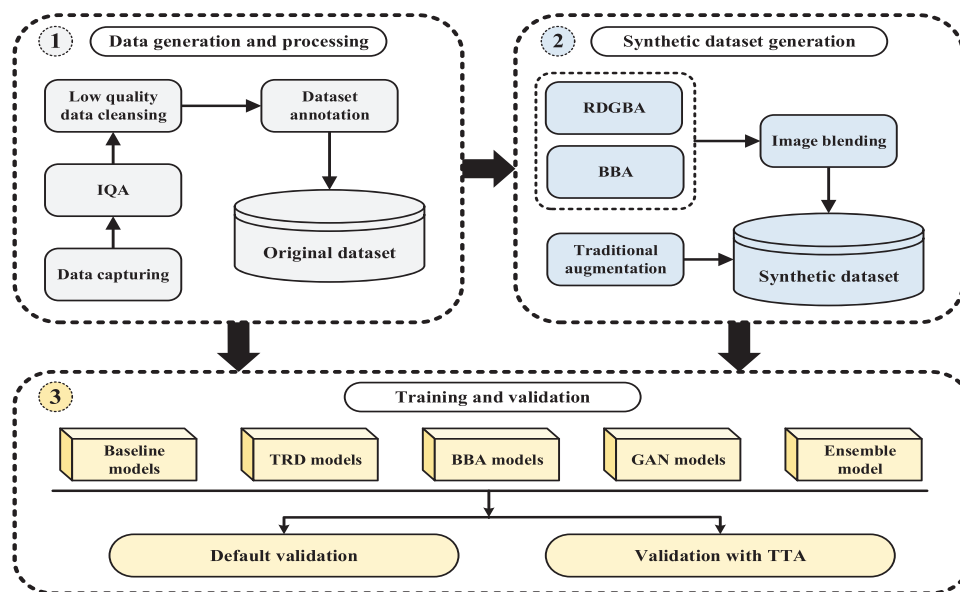


FIGURE 5 The block diagram of the proposed method.

TABLE 4 Evaluation results of baseline models for object detection.

Model name	mAP	F1-score	Speed (fps)
YOLOv5n	0.35	0.34	110
Faster-RCNN-R50-C4	0.42	0.41	5
YOLOv5n	0.37	0.34	22

These results were obtained by training the three models of YOLOv5n, Faster-RCNN-R50-C4, and YOLOR-W6 with 1000 images for 50 epochs. 80% of these images were used for training and 20% for validation. Considering the speed of the YOLOv5 model and its similar accuracy to the YOLOR model, YOLOv5 has been used for developing the detectors in this paper.

To create a baseline model, 25,000 images from the IRRDD were used for training. First, the data were divided with 80/20 ratio into two portions for training and validation, respectively. The three considered models were trained with the default hyper-parameters. Based on the number of parameters, the models were created in three sizes (N (nano), M (medium), and X (extra-large)). The initial training parameters of these models have been presented in Table 5. The NVIDIA Tesla T4 system was employed to train these models.

4.2 | Data augmentation

In this section, we will explore the data augmentation techniques that have been implemented to improve the precision of the baseline models. This section is divided into the two subsections of traditional and complex data augmentation. In the former subsection, we will survey the data augmentation techniques that are employed in the majority of research works. In

TABLE 5 Training parameters.

Model name	IRDn	IRDm	IRDx
Epochs	100	100	100
Lr0	0.0032	0.0032	0.0032
Lr1	0.12	0.12	0.12
Parameters (M)	1.9	21.2	86.7
Image size	640	640	640
Batch size	32	32	8
Optimizer	SGD	SGD	SGD

the latter subsection, the road damage GAN based augmentation (RDGBA) and the BBA methods, which are used in this research, will be examined.

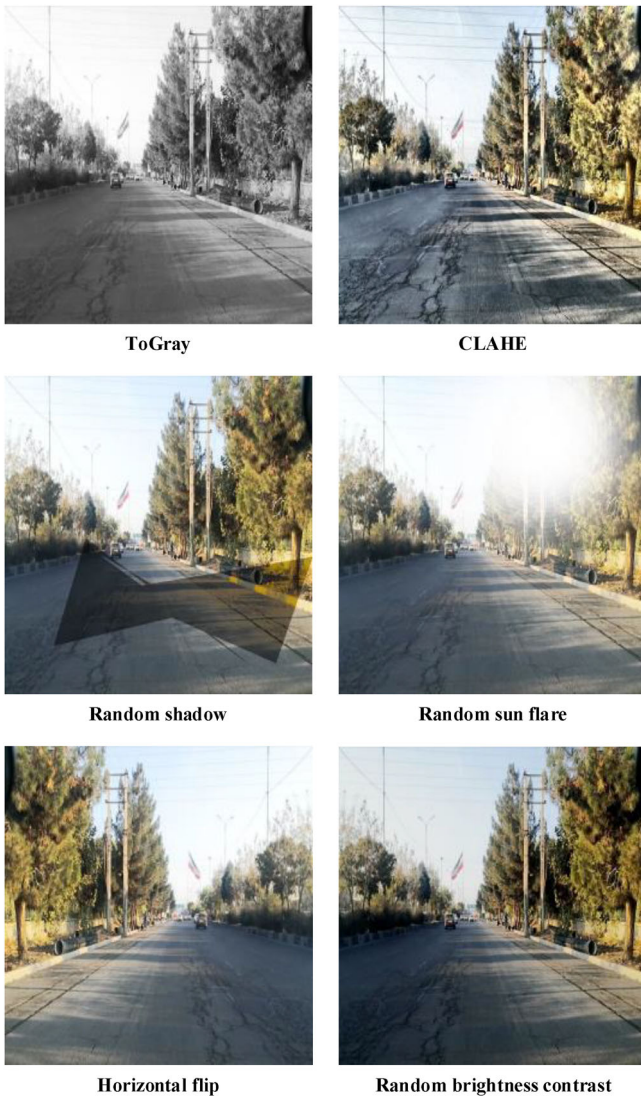
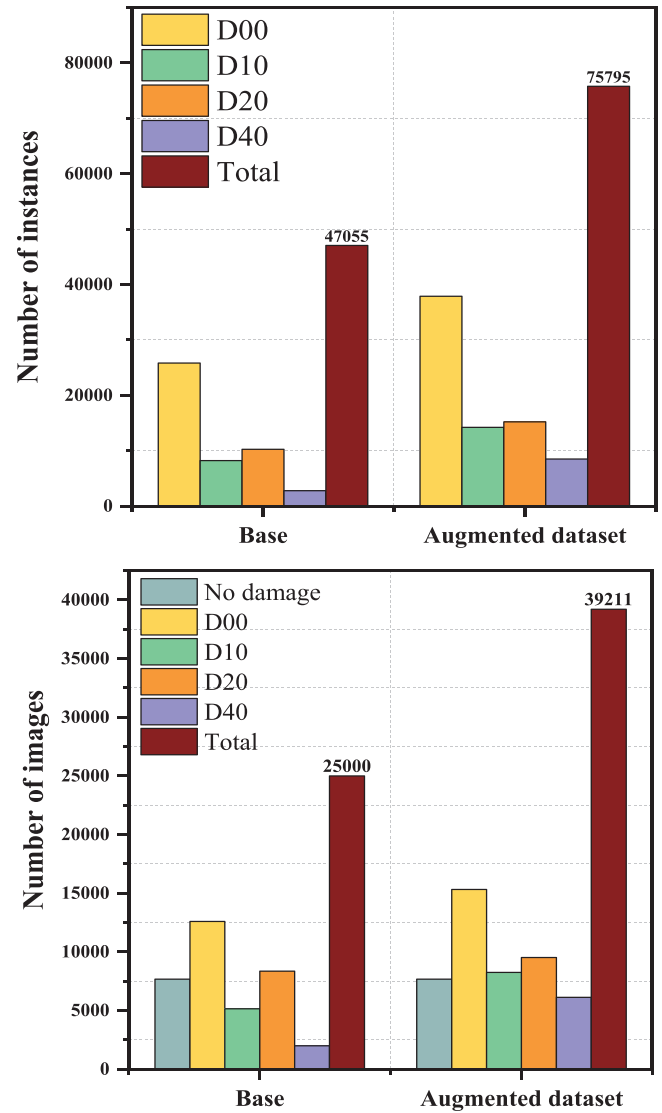
4.2.1 | Traditional data augmentation

In this research, to increase the detection precision and to enhance the generalization of the models, several methods are employed for augmenting the available data and improving the classes that have not been distributed properly. All the traditional augmentations and their parameters are listed in Table 6. These augmentations are created by using the Albumentation [73] tool and placed in the training pipeline. The extra data generated by the augmentation techniques such as random shadow, random sun flare, and random brightness contrast help boost the accuracy and the generalization of the models under realistic conditions. Figure 6 shows the various types of traditional augmentations.

First, the existing images are classified based on the number of images in different classes. Since the various types of

TABLE 6 Traditional data augmentation list.

Type of augmentation	Augmentation parameters
Contrast limited adaptive histogram equalization (CLAHE)	clip_limit = 4.0, tile_grid_size = (8, 8)
Random sun flare	flare_roi = (0, 0, 1, 0.5), angle_lower = 0, angle_upper = 1, num_flare_circles_lower = 6, num_flare_circles_upper = 10, src_radius = 250, src_colour = (255, 255, 255)
Random brightness contrast	brightness_limit = 0.2, contrast_limit = 0.2, brightness_by_max = True, always_apply = False
Random shadow	shadow_roi = (0, 0.5, 1, 1), num_shadows_lower = 1, num_shadows_upper = 2, shadow_dimension = 5

**FIGURE 6** Traditional data augmentation samples.**FIGURE 7** (a) Number of images per class (b) Number of instances per class.

road damages do not have uniform distributions and the D10 and D40 classes have fewer instances than the other classes, the horizontal flip, CLAHE, and ToGray augmentation techniques are carried out for all the images in these classes. Data augmentation techniques such as CLAHE enhance the precision of damage detectors. This method has been employed in ref. [74] to improve the precision of crack segmentation and detection. Figure 7 shows the distributions of the number of images per class and the number of instances following the abovementioned augmentation procedures. Using these data, the IRDn model is trained for 150 epochs.

4.2.2 | Complex augmentation: bounding box augmentation

In this section, we first describe the BBA method, which is employed to increase the number of images in the D10 and

ALGORITHM 1 Road damage BBA.

Require: Road damage dataset with annotations $RD (Img_i, Lbl_i)$
 $(i = 1, \dots, n)$
 In the dataset $Aclass$: represents the area of a object class in an image.
 n represents number of images in the dataset
Inputs: Img_i and Lbl_i , Augmentation type $AugType$,
 Augmentation class $AugClass$
Output: Bounding box augmented Images I_{BBA}

```

01 First Step: BBA
02 For  $i_0 < n$  :
03   If  $AugClass$  is available in  $Lbl_{i_0}$  :
04      $I_{BBA0}(i_0) = \text{Flip}(AugType) \text{ the } Aclass_{i_0}$ 
05   End If
06 End For

```

D40 classes. In the BBA technique, instead of applying the considered changes to a whole image, the image transformations are applied to the bounding box of a specific object. For example an object could undergo horizontal and vertical flips within its bounding box. This requires an annotated dataset. In this approach, some transformations may produce noticeable differences in the images. The initial data in this case are generated by means of the Algorithm 1. In this algorithm, first, the considered class and augmentation type are selected. Then by using the available data, the BBA process begins. For an initial evaluation of this method, the data have only been subjected to the following BBA types: BBA-horizontal flip, BBA-vertical flip, and the BBA-horizontal/vertical flip.

Because of the difference in their backgrounds and textures, the data generated in this way are not suitable for training the damage detector models. These data cause the detectors to learn the kind of features that eventually reduce the accuracy of these detectors. For example, the edges of the bounding box hori-

ALGORITHM 2 Automatic mask generation and modified Poisson blending.

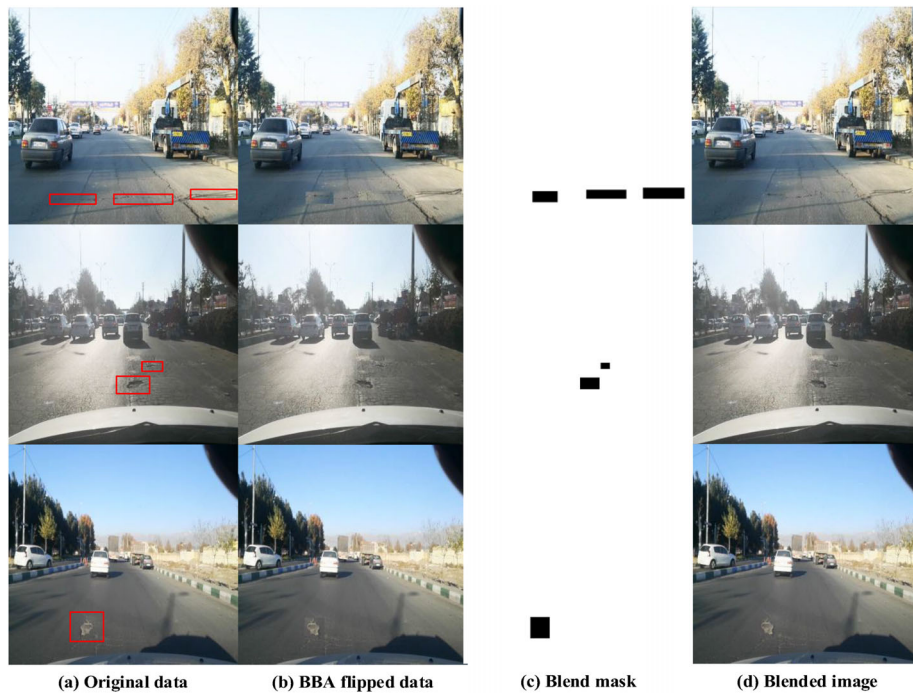
Require: Augmented Dataset $ARD (AImg_i, ALbl_i)$,
 Original Dataset $RD (Img_i, Lbl_i)$ $(i = 1, \dots, n)$
 In the dataset $Aclass$ represents the area of a object class in an image.
 n represents number of augmented images
Inputs: AMg_i , Img_i , Lbl_i , Augmentation class $AugClass$
Output: Fully augmented images and labels $AMg2$, $ALbl2$
First Step: Mask generation and MPB

```

01 For  $i < n$  :
02   If  $Lbl_i$  Equals  $AugClass$  :
03      $M_i$  = Create a white image in the size of  $Img_i$ 
04      $Aclass_i$  =  $0.75 \times Aclass_i$ 
05     // Do this for a label that is equal  $AugClass$ 
06     = Fill  $Aclass_i$  in reference  $Lbl_i$  position
07     with black pixels in  $M_i$ 
08      $ALbl2$  = Generate new  $Lbl_i$  based on the new size of  $Aclass_i$ 
09      $I_{Source}$  =  $AMg_i$ 
10      $I_{Target}$  =  $Img_i$ 
11      $I_{Mask}$  =  $M_i$ 
12      $AMg2$  = MPB (  $I_{Source}$ ,  $I_{Target}$ ,  $I_{Mask}$  )
13   End If
14 End For
15 Second Step: Quality control
16 Inspect each generated image and delete low-quality images

```

zontal flipped images are more prominent than the edges of the original images. To combine an image with a background image, the modified Poisson blending technique has been adopted [19]. This algorithm blends two images so that the gradient between them remains proportional. In this approach, each image needs a mask, which defines the blending territory. In most research works, this mask is created manually. Here, we produce a mask automatically by means of the proposed Algorithm 2. In this algorithm, by using the annotations related to the augmented classes, the bounding boxes of these classes are reduced to some extent and a mask is created for one considered class

**FIGURE 8** BBA samples.

only. Finally, by using the image masks and the bounding box augmented images, the modified Poisson blending operation is carried out and new images are produced. The quality of the new data is checked manually, and the data which have not been augmented properly are discarded. Because of the large-size images, the D20 class produces unrealistic images and it is not suitable for BBA-flip operations; so, the data generated for this class have not been used for model training. In order to examine the performance of this method, the IRDn model was trained with the new data for 150 epochs. The BBA results are illustrated in Figure 8.

4.2.3 | Complex augmentation: road damage GAN based augmentation

It is difficult to collect a set of road damage data for just one type of road damage class. A lot of data will be needed to train a robust damage detection model. In the road damage GAN based augmentation method, the data containing no damages are used to generate the new data with different road damages. This technique requires a dataset of various damages and an algorithm for combining the damage-free data with the road damage samples. This data augmentation mechanism has the following steps:

Separating all the pictures that contain road damages objects from the images of the IRRDD in order to form the datasets needed by the GANs

1. Training GAN models for producing the images with synthetic damages
2. Collecting the damage-free data
3. Blending the road damages into the damage-free images by using the automatic road damage blender (ARDB)
4. Checking the quality of the produced images and eliminating the low-quality data
5. Using the obtained data to train the damage detection models

Before applying this data augmentation procedure, first, all the road damages shown in the IRRDD were grouped as separate images in corresponding classes. Then, for training GAN models, these images were converted to square sizes. Table 7 shows the number of images in each class and the transformed sizes of these images. In order to increase the number of the training samples in the GAN models, the images in the damage classes of D10 and D40 underwent the flip and the brightness data augmentation procedures.

The GANs are employed in many research works for data augmentation and for enhancing the accuracy of the detectors [75]. By using these networks, new training data can be generated for the object detection models. Every GAN comprises a generator section and a discriminator section, which are constantly competing with each other. The discriminator checks the quality of the data produced in the generator. The generator unit always tries to produce real data and the discriminator unit always tries to detect fake data. To be able to create appro-

TABLE 7 A list of the training datasets for the road damage detection models.

Dataset	Number of images	Number of instances	Size of images (pixels)
IRRDD-Baseline	25000	47055	640 × 640
IRRDD-Augmented	39211	75795	640 × 640
IRRDD-BBA-D10	43457	67233	640 × 640
IRRDD-BBA-D40	41329	57247	640 × 640
IRRDD-BBA-D10D40	45127	68927	640 × 640
IRRDD-D40-RDGBA-25	39401	56156	640 × 640
IRRDD-D40-RDGBA-50	39557	56759	640 × 640
IRRDD-D40-RDGBA-75	39871	57362	640 × 640
IRRDD-D40-RDGBA-100	39997	59553	640 × 640
IRRDD-D10-RDGBA-25	40175	57578	640 × 640
IRRDD-D10-RDGBA-50	40524	59603	640 × 640
IRRDD-D10-RDGBA-75	41960	61628	640 × 640
IRRDD-D10D40-RDGBA-50	40881	62834	640 × 640
GAN-D10	8251	—	96 × 96
GAN-D40	6120	—	64 × 64

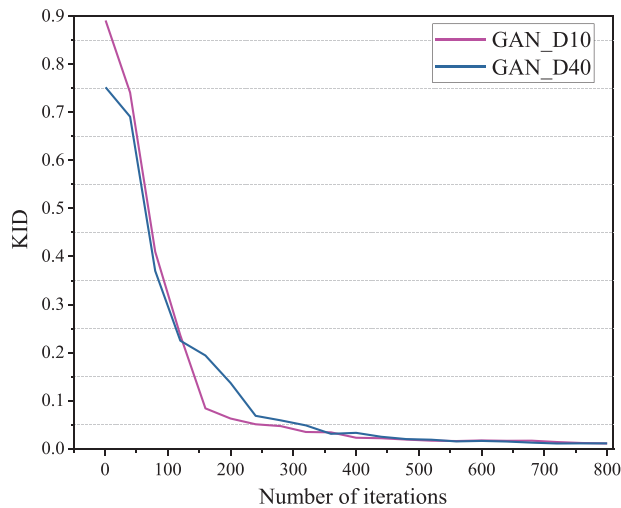
priate models, a large volume of data has to be supplied to the GANs. An adaptive discriminator augmentation mechanism (StyleGAN2-ADA) [21] is employed to create strong generator models with limited data. In this research, two GAN models have been established for the D10 and D40 damage classes. The performances of the GAN_D40 and GAN_D10 models in this paper have been evaluated by applying the kernel inception distance (KID) criterion, which is lighter computationally and more stable numerically. Each of these GAN models has been trained by 2500k images. Training the models with this number of images has provided adequate synthetic sample data of sufficient diversity to be used in the RDGBA technique. The performance results of these models have been highlighted in Figure 9. The training results of the GAN_D40 and GAN_D10 models are illustrated in Figures 10 and 11, respectively. By using each of these models, 5000 new damages from corresponding classes were produced. In the annotation phase of the IRRDD, many of the images contained no road damage. These images were used by ARDB to generate new training data.

The architecture of ARDB algorithm is illustrated in Figure 12. The ARDB method comprises three main steps. In the first step, by using the data with no damages and with a variety of damages, and by considering the number of samples, the damages are randomly pasted onto the images within a specific bounding box. Considering the image sizes, the selected bounding box is between the pixel bounding boxes of ($y_{min} = 380$, $y_{max} = 520$) and ($x_{min} = 120$, $x_{max} = 520$). In most of the images in the IRRDD, this bounding box includes the roads.

The images with the pasted road damages have certain edges which cause the models to learn the wrong features. In the second step, these images are blended by using the Algorithm

TABLE 8 The computational costs and time complexities of the models introduced in the paper.

Model name	Pre-process (ms)	Inference (ms)	NMS (ms)	Total time (ms)	FLOPS (B)
IRDn	0.4	8	1.3	9.7	4.2
IRDn_TTA	0.4	24	1.8	26.2	4.2
IRDn_AugTrd_2	0.4	8.4	1.3	10.1	4.2
IRDn_AugTrd_2_TTA	0.4	26	1.7	28.1	4.2
IRDn_BBA_D10D40	0.4	8.7	1.3	10.4	4.2
IRDn_BBA_D10D40_TTA	0.4	25	1.8	27.2	4.2
IRDn_D10D40_RDGBA50	0.4	8.9	1.3	10.6	4.2
IRDn_D10D40_RDGBA50_TTA	0.4	29	1.8	31.2	4.2
IRDm	0.6	40	1.4	42	48.3
IRDm_TTA	0.6	73	1.3	74.9	48.3
IRDx	0.6	70	1.4	72	204.7
IRDx_TTA	1	143	1.4	145.4	204.7
IRDn_Ensemble	0.5	35	2.5	38	12.6
IRDn_Ensemble_TTA	0.4	87	7.1	94.5	12.6

**FIGURE 9** The performance results of the GAN_D40 and GAN_D10 models based on the KID criterion.

2 and the reference images containing no damage. The data generated by this algorithm enjoy a higher quality; however, due to the perspective distortion, some samples do not have an adequate quality. In the final step, all the produced images were reexamined in order to eliminate the low-quality specimens. To evaluate the performance of this method, the “IRDn” model was trained with the new data for 200 epochs. Some image samples created by the RDGBA approach are displayed in Figure 13.

All the datasets used to train the models of this paper have been presented in Table 8. The IRRDD-Baseline dataset has been used to train the baseline detection models. In the road damage classes, the IRRDD-Augmented dataset provides a more uniform distribution than the IRRDD-Baseline dataset. The IRRDD-BBA-D10, IRRDD-BBA-D40, and the IRRDD-

BBA-D10D40 dataset have been employed to train and test the BBA-based models. Using the RDGBA technique, eight datasets have been created to test and improve this method. The numbers at the end of the names of these datasets indicate the ratio of the number of synthetic data generated by means of the RDGBA technique to the number of real data in the reference dataset.

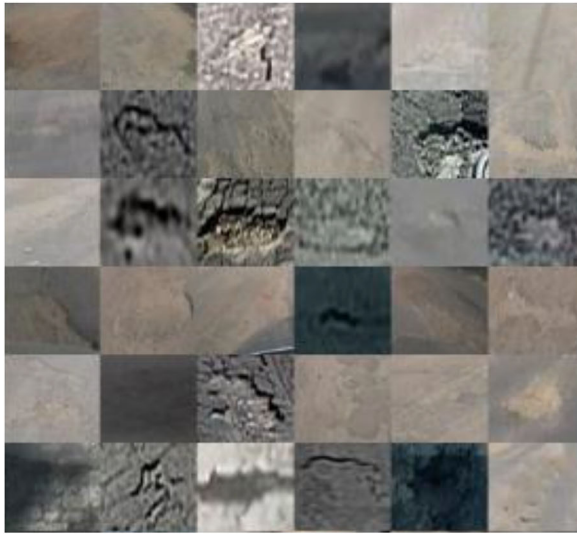
5 | ANALYSIS OF RESULTS

In this section, the different types of data augmentation methods (traditional, complex) are evaluated and the results of these models are compared.

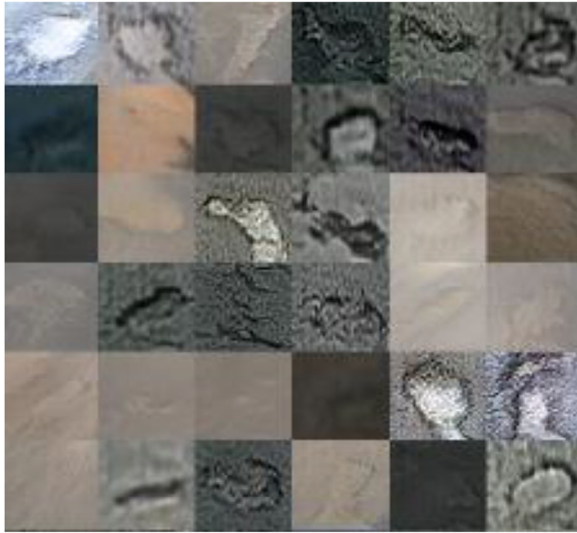
5.1 | Measuring parameters

We use F1-score and mAP to assess the performance of these models in the IRRDD. Here, a damage detection is considered to be correct when the proposed bounding box of a model contains at least 50% of the ground truth bounding box. This criterion is known as the intersection over union (IoU). The precision measure (Eq. (1)) shows the ratio of the number of true positive detections to the total number of detections considered as positive. In Eq. (1), true positive means the correct detection of a road damage with an IoU > 0.50. Also, false positive indicates the incorrect detection of a road damage in a bounding box in which there is no type of road damage. The recall measure (Eq. (2)) shows the ratio of the number of true positive detections to the total actual positive detections.

$$\text{Precision} = \frac{\text{True positive}}{\text{True positive} + \text{False positive}} \quad (1)$$



(a) Real data



(b) GAN generated data

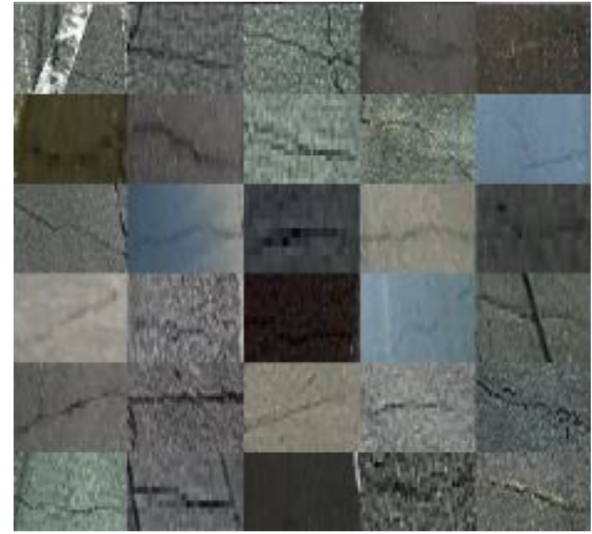
FIGURE 10 Real D40 versus GAN_D40 images.

$$\text{Recall} = \frac{\text{True positive}}{\text{True positive} + \text{False negative}} \quad (2)$$

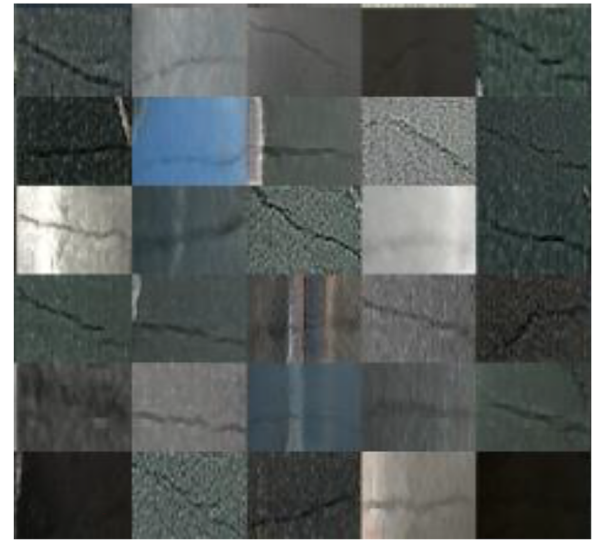
The F1-score strikes a balance between the precision and the recall measures. This metric computes the overall accuracy of a model via Equation (3).

$$\text{F1 - score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (3)$$

Another criterion used in this research to evaluate the models is the mAP value. This metric measures the mean precision for all the damage classes. The mAP is computed for the recall values from 0 to 1. In this paper, the average mAP values for 10 different IoUs (between 0.5 and 0.95, with a step of 0.5) have



(a) Real data



(b) GAN generated data

FIGURE 11 Real D10 versus GAN_D10 images.

also been computed. The results of the considered models and their trainings, data explorations etc. have been listed in Table 9.

5.2 | Baseline models

The training results of the three considered models (IRDn, IRDm, and IRDx) show that, with regards to the number of parameters involved, the IRDx model achieves the highest detection accuracy. After it is trained for 100 epochs, this model reaches a mAP of 0.515 and F1-score of 0.519. However, comparing the inference speeds the baseline models in Table 8 shows that this model has the lowest inference speed among the said models. The IRDn model, which is the smallest of the mentioned models, is considered to be the best model for real-time applications. Thus, the IRDn is chosen as the

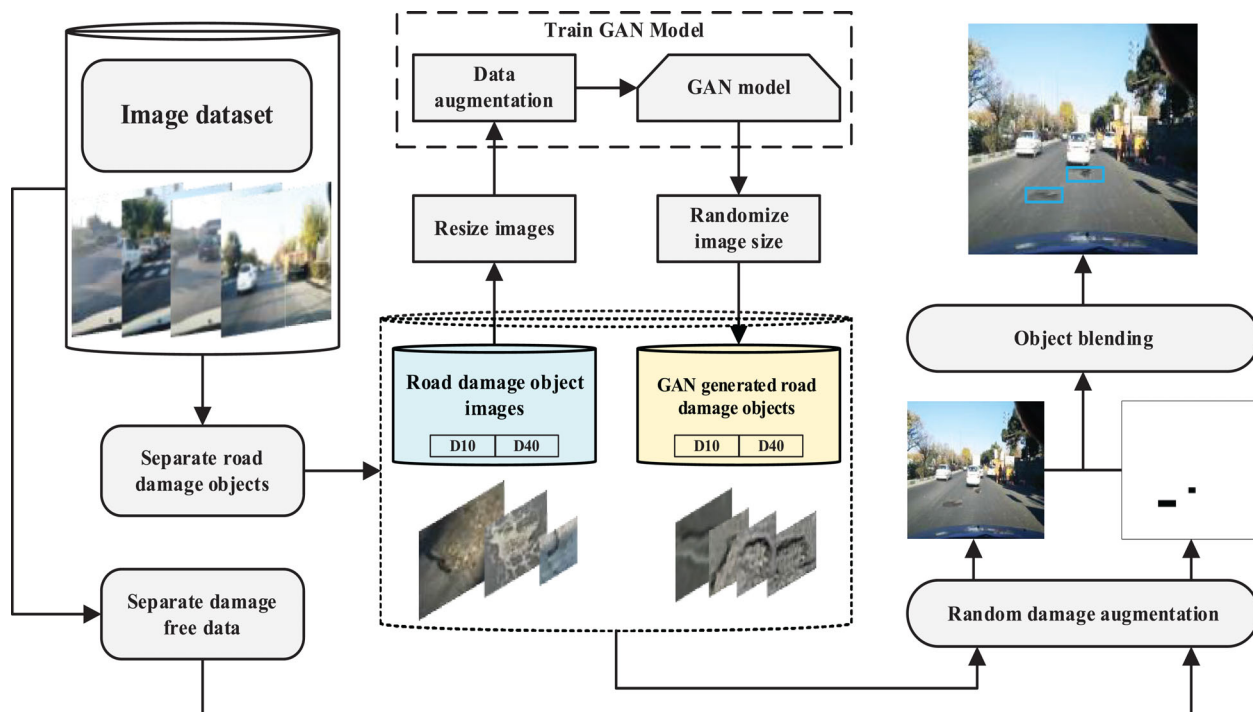


FIGURE 12 The ARDB architecture.

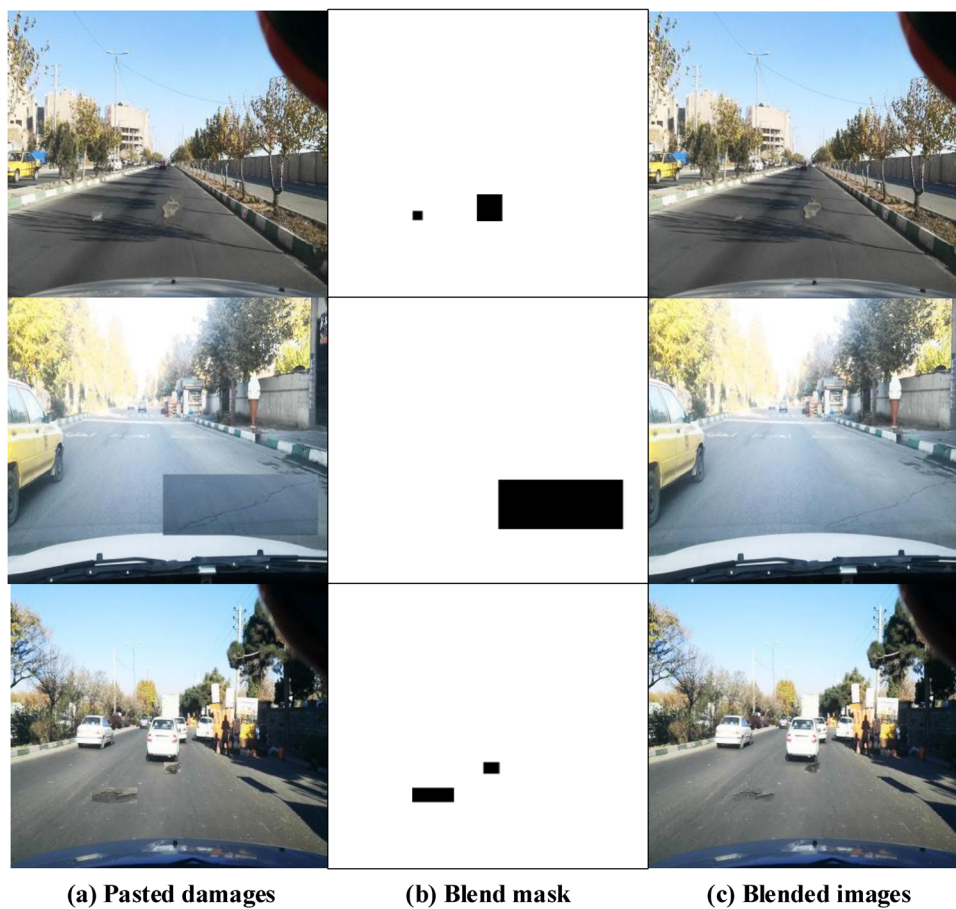


FIGURE 13 RDGBA samples.

TABLE 9 Validation metrics for the all models.

Model name	F1-score					mAP IoU > 0.5					mAP IoU 0.5–0.95
	D00	D10	D20	D40	Overall	D00	D10	D20	D40	Overall	
IRDn	0.552	0.38	0.611	0.375	0.488	0.551	0.4	0.623	0.283	0.464	0.176
IRDn_TTA	0.561	0.42	0.633	0.384	0.508	0.561	0.404	0.67	0.287	0.481	0.181
IRDm	0.567	0.45	0.638	0.379	0.508	0.579	0.483	0.614	0.341	0.505	0.205
IRDm_TTA	0.586	0.48	0.641	0.384	0.522	0.58	0.51	0.623	0.363	0.519	0.215
IRDx	0.571	0.51	0.616	0.379	0.519	0.59	0.51	0.607	0.352	0.515	0.218
IRDx_TTA	0.59	0.53	0.593	0.384	0.525	0.599	0.535	0.632	0.373	0.535	0.235
IRDn_AugTrd_1	0.561	0.482	0.617	0.383	0.512	0.577	0.407	0.637	0.315	0.484	0.185
IRDn_AugTrd_1_TTA	0.564	0.46	0.611	0.393	0.509	0.57	0.399	0.649	0.336	0.489	0.187
IRDn_AugTrd_2	0.562	0.481	0.615	0.389	0.512	0.582	0.426	0.657	0.311	0.494	0.194
IRDn_AugTrd_2_TTA	0.567	0.46	0.631	0.39	0.512	0.572	0.422	0.681	0.311	0.497	0.196
IRDn_AugTrd_3	0.565	0.427	0.624	0.319	0.489	0.57	0.415	0.63	0.252	0.467	0.171
IRDn_AugTrd_3_TTA	0.568	0.422	0.63	0.365	0.5	0.574	0.413	0.655	0.284	0.481	0.185
IRDn_BBA_D40	0.571	0.473	0.626	0.391	0.521	0.574	0.452	0.661	0.332	0.505	0.202
IRDn_BBA_D40_TTA	0.559	0.469	0.639	0.349	0.511	0.559	0.464	0.676	0.319	0.505	0.204
IRDn_BBA_D10	0.527	0.5	0.602	0.36	0.502	0.533	0.475	0.628	0.34	0.494	0.196
IRDn_BBA_D10_TTA	0.504	0.475	0.651	0.379	0.51	0.526	0.482	0.663	0.338	0.502	0.201
IRDn_BBA_D10D40	0.564	0.477	0.624	0.406	0.521	0.573	0.444	0.648	0.332	0.499	0.2
IRDn_BBA_D10D40_TTA	0.576	0.442	0.649	0.368	0.518	0.582	0.45	0.673	0.308	0.503	0.201
IRDn_D40_RDGBA25	0.559	0.471	0.617	0.381	0.507	0.571	0.407	0.631	0.312	0.481	0.186
IRDn_D40_RDGBA25_TTA	0.532	0.455	0.611	0.367	0.492	0.564	0.399	0.642	0.331	0.484	0.188
IRDn_D40_RDGBA50	0.564	0.463	0.63	0.407	0.521	0.54	0.402	0.656	0.331	0.483	0.185
IRDn_D40_RDGBA50_TTA	0.572	0.474	0.641	0.366	0.518	0.576	0.424	0.678	0.339	0.505	0.204
IRDn_D40_RDGBA75	0.56	0.434	0.617	0.408	0.505	0.567	0.406	0.659	0.357	0.498	0.202
IRDn_D40_RDGBA75_TTA	0.55	0.439	0.64	0.37	0.5	0.563	0.416	0.692	0.329	0.5	0.2
IRDn_D10_RDGBA25	0.534	0.478	0.635	0.375	0.506	0.569	0.402	0.651	0.331	0.489	0.189
IRDn_D10_RDGBA25_TTA	0.515	0.453	0.651	0.368	0.497	0.576	0.424	0.675	0.321	0.499	0.201
IRDn_D10_RDGBA50	0.56	0.452	0.633	0.402	0.512	0.568	0.407	0.651	0.332	0.49	0.193
IRDn_D10_RDGBA50_TTA	0.548	0.471	0.638	0.361	0.505	0.576	0.423	0.68	0.338	0.505	0.203
IRDn_D10_RDGBA75	0.56	0.453	0.636	0.385	0.509	0.563	0.41	0.658	0.313	0.486	0.186
IRDn_D10_RDGBA75_TTA	0.563	0.45	0.641	0.356	0.503	0.569	0.407	0.67	0.313	0.49	0.201
IRDn_D10D40_RDGBA50	0.564	0.475	0.64	0.421	0.525	0.575	0.476	0.67	0.346	0.517	0.216
IRDn_D10D40_RDGBA50_TTA	0.57	0.48	0.66	0.356	0.517	0.562	0.445	0.61	0.349	0.492	0.195
IRDn_Ensemble	0.559	0.491	0.631	0.364	0.512	0.571	0.44	0.663	0.362	0.509	0.207
IRDn_Ensemble_TTA	0.575	0.469	0.66	0.39	0.525	0.585	0.456	0.689	0.381	0.528	0.225

baseline model for implementing the augmentation procedures. All the experiments in this research are conducted in two ways. In the first case, the validation step is carried out without the TTA process, and in the second case, it is implemented with the TTA. In the TTA process, which involves a heavier processing load, the image sizes roughly get 30% larger and then the images undergo the flip operation. Then the detection results of the models for these augmentations are merged together before conducting the non-maximum suppression (NMS) pro-

cedures. The results of these models are presented in Table 9 and Figure 14. These results indicate that with the implementation of TTA, all the baseline models have a better performance. However, the TTA approach doubles the processing time of the models. The results of the baseline models show that the IRDx model achieves the best detection performance, but has the lowest inference speed. Also, the IRDm and IRDn models are not much different in terms of their overall mAP and F1-score values.

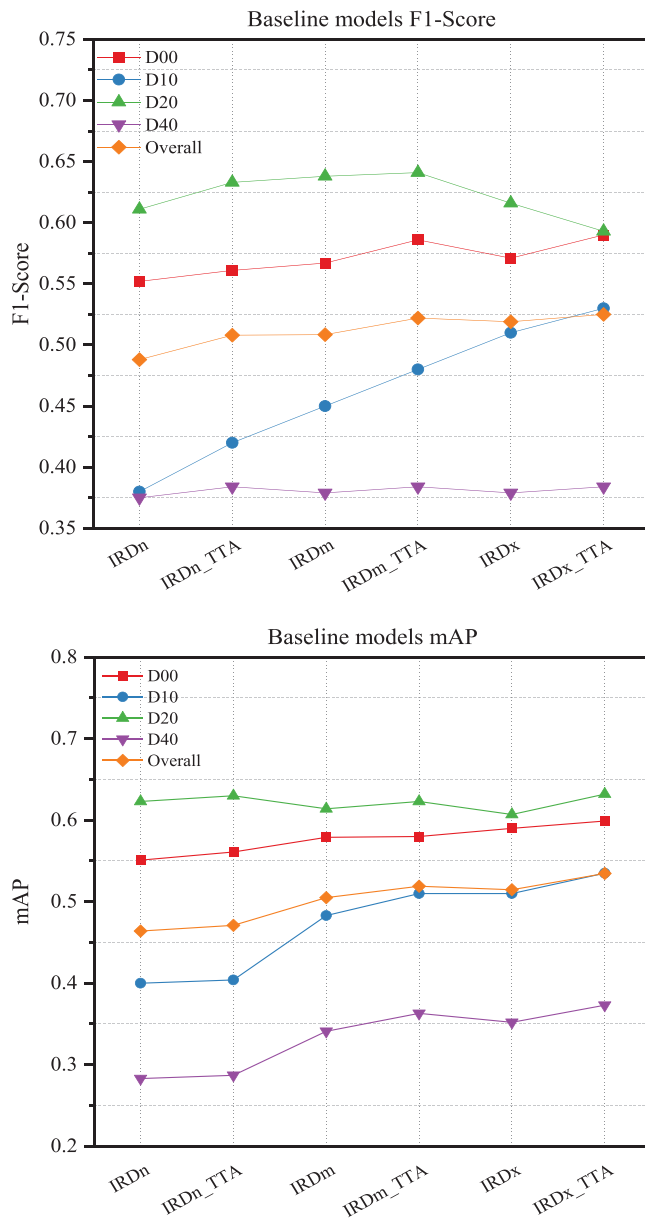


FIGURE 14 Baseline models' mAP and F1-score.

5.3 | Traditional augmentation results

The three considered models are trained by applying the traditional augmentation procedures. First, the augmented data of the D10 and D40 classes are added separately to the main dataset. Then by using these data, the detector models are trained separately. The new data are the outcomes of the horizontal flip, CLAHE, and the ToGray augmentation operations. Each model is trained by means of different augmentation parameters. The “IRDn_AugTrd_1” model was trained by using all the augmentations and the probability of 0.05. After training this model for 150 epochs, the obtained results show no tangible change in the values of mAP and F1-score. The IRDn_AugTrd_2 model was trained with the probability of 0.1. Relative to the reference model (IRDn),

the F1-score and mAP of this model is improved by 4.92% and 7.11%, respectively. By changing the brightness level of some test data, it was demonstrated that the reference model is very sensitive to these variations. Training the models with these augmented data enhances the generalization and robustness of the object detectors. The augmentation probability of the IRDn_AugTrd_3 model is 0.2. The performance of this model with respect to the original validation data is weaker than that of the “IRDn_AugTrd_2” model. The F1-score of this model is 0.489. Increasing the probability of this augmentation could make the model biased under unrealistic conditions, which reduces the model's accuracy in the validation data. The results of these models have been compared in Table 9 and Figure 15. The findings of this section shows that augmentation probability of 0.1, improves generalization of the road damage detection models.

In the RDGBA technique, first, synthetic data were generated for the D40 class. These data consisted of 1000 new images along with 2000 D40 instances. These data were reexamined and 381 low-quality images were omitted from this set. Next, by applying the horizontal flip, CLAHE, and the ToGray techniques these data were augmented and their number was increased to 4000 images. The new data were then added to the reference dataset and the “IRDn” model was trained with the new dataset for 200 epochs.

The training results of this model indicate the diminishing of its F1-score with regards to the test data obtained in both cases of with and without TTA. The number of annotations of the D40 class is very low. Using a large volume of new data has led to the diminishing of the F1-score and mAP. The trained model has become biased with respect to the synthetic data. To prevent the detection model from becoming biased in the D40 class, synthetic data should be added to real data in balanced proportions. In order to find out the right proportions of data, three tests were devised in which the synthetic data were added to the training data in the proportions of 25, 50 and 75% of the number of real data annotations. Then, considering the percentage of the RDGBA for the D40 class, three models of IRDn_D40_RDGBA25, IRDn_D40_RDGBA50, and IRDn_D40_RDGBA75 were trained with the new data for 200 epochs. The training results of these models are listed in Table 9.

The training results indicate that the IRDn_D40_RDGBA50 model has the best performance among the three models trained by this method. This model has achieved a F1-score of 0.521. Also, the F1-score and the mAP value of this model in the D40 class are 8.53 and 17% higher than those of the baseline model.

Next, to find out the performance of this method in the D10 damage class, synthetic data were generated. These data were added to the training dataset in the proportions of 25%, 50%, and 75% of the number of original data annotations in the D10 class. Then, considering the percentage of the RDGBA for the D10 class, three models of “IRDn_D10_RDGBA25”, “IRDn_D10_RDGBA50”, and “IRDn_D10_RDGBA75” were trained with the new data for 200 epochs. The training results of these models are presented in Table 9.

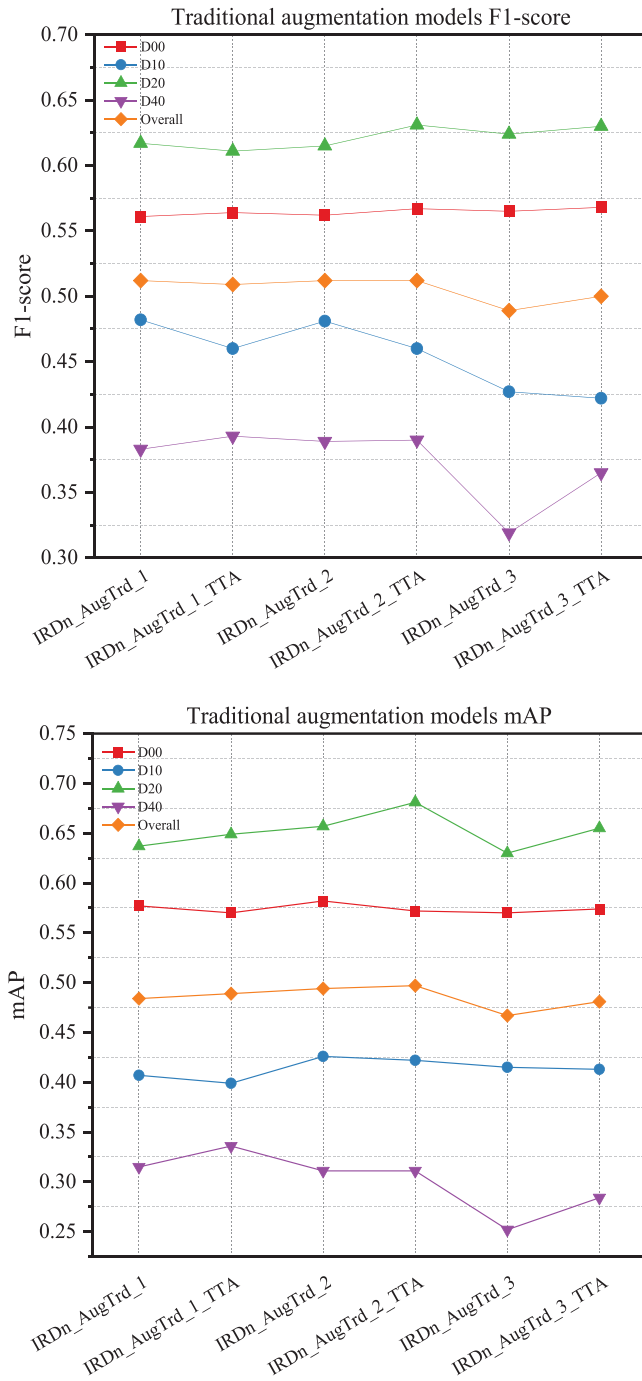


FIGURE 15 Traditional augmentation models' mAP and F1-score.

5.4 | Complex augmentation results

In this section, 11 models will be evaluated. First, to investigate the effect of BBA, this augmentation scheme was applied separately on the D10 and D40 damage classes. One model for the D10 class and one model for the D40 class were trained for 150 epochs. In these trainings, traditional augmentations with the probability of 0.1 were applied. In the “IRDn_BBA_D40” model, the F1-score was improved by 4.27% in the D40 class. Also, in the “IRDn_BBA_D10” model, the F1-score improved

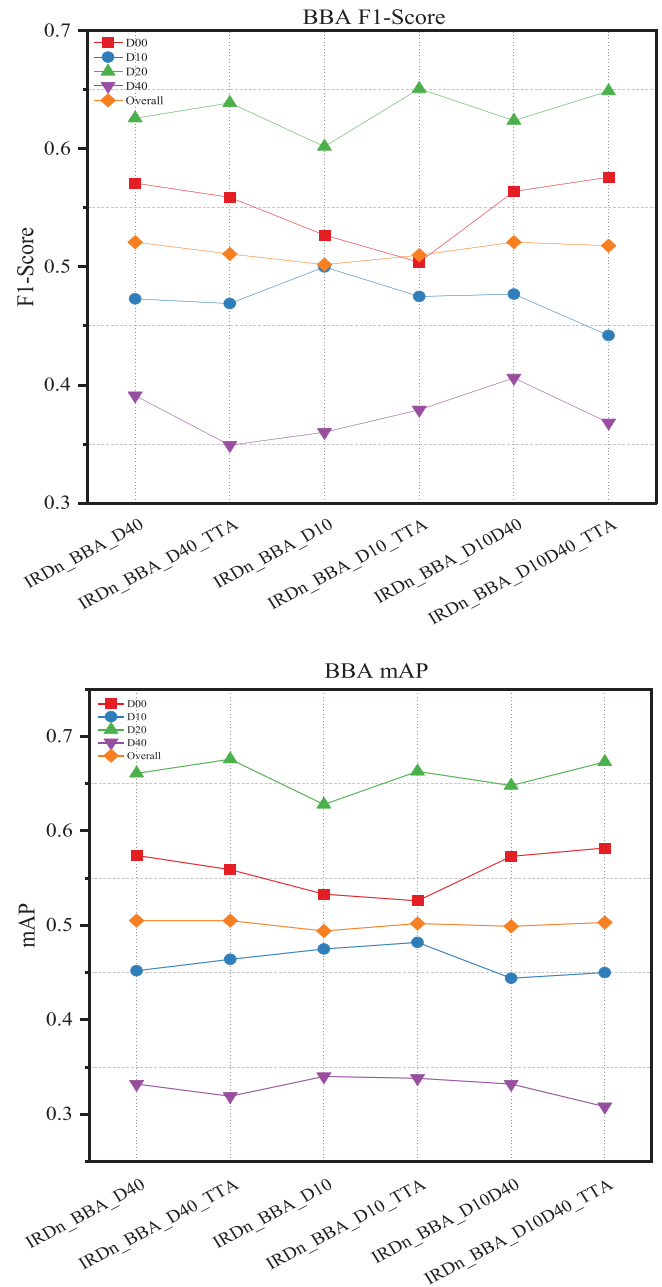


FIGURE 16 The mAP and F1-score values for the BBA models.

by 31.58% in the D10 class. Then, in view of the obtained results, another model was trained for 150 epochs by using all the new data. The F1-score of this model increased by 6.76% over the reference model. Also, the mAP value of this model was elevated by 7.54%. The results of these models have been compared in Table 9 and Figure 16. These results show that the BBA approach can enhance the performance of the road damage detectors in the classes with limited data.

The training results show that the “IRDn_D10_RDGBA50” model has the best performance among the three models trained by this technique. This model achieves a F1-score of 0.512. Also, the F1-score and the mAP value of this model in

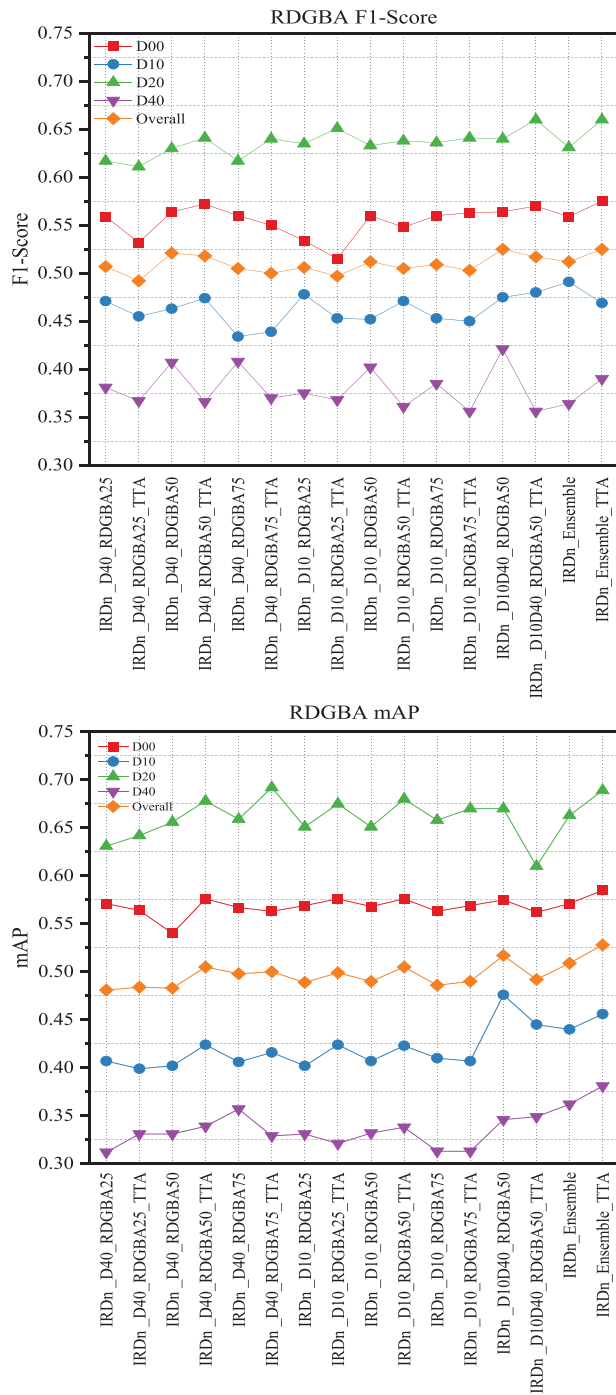


FIGURE 17 The mAP and F1-score values for the complex augmentation models.

the D10 class are 19 and 1.75% higher than those of the baseline model, respectively.

Subsequently, by using 50% of the synthetic data in the D10 and D40 classes, a model was trained for 200 epochs. This model is stronger than the models that are trained separately. The F1-score and the mAP value of this model are 0.525 and 0.517, respectively, which are 7.58% and 11.42% greater than those of the baseline model. The results of this section reveals that the models trained by synthetic data can detect the

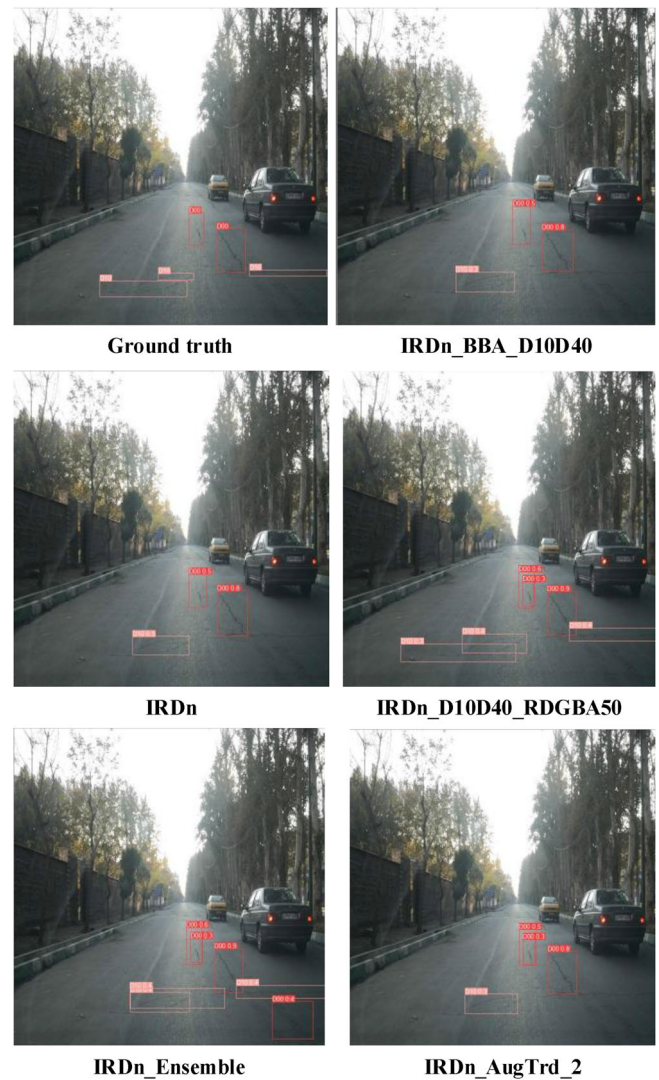


FIGURE 18 Detection samples obtained by different trained models in IRRDD.

road damages more effectively. This greater capability is influenced by the amount of data augmentation implemented in this approach. Using a large number of synthetic images makes the models biased under unrealistic conditions, which reduces the precision and the recall of the models. Many of the produced synthetic data have low qualities and should not be used for model training. The results of these models are presented in Table 9 and Figure 17.

Eventually, the models of “IRDn_D10D40_RDGBA50”, “IRDn_BBA_D10D40”, and “IRDn_AugTrd_2” were combined together to form the “IRDn_Ensemble” model. These models have had the best performance in each of the test groups. All the considered models participate in detection operations with equal weights. Each of these models is trained by focusing on a particular data augmentation technique. This process helps produce a precise and robust model in all the test conditions considered. The processing times for the baseline models in Table 8 show the shortest detection time for the IRDn model and the longest time for the IRDx model.

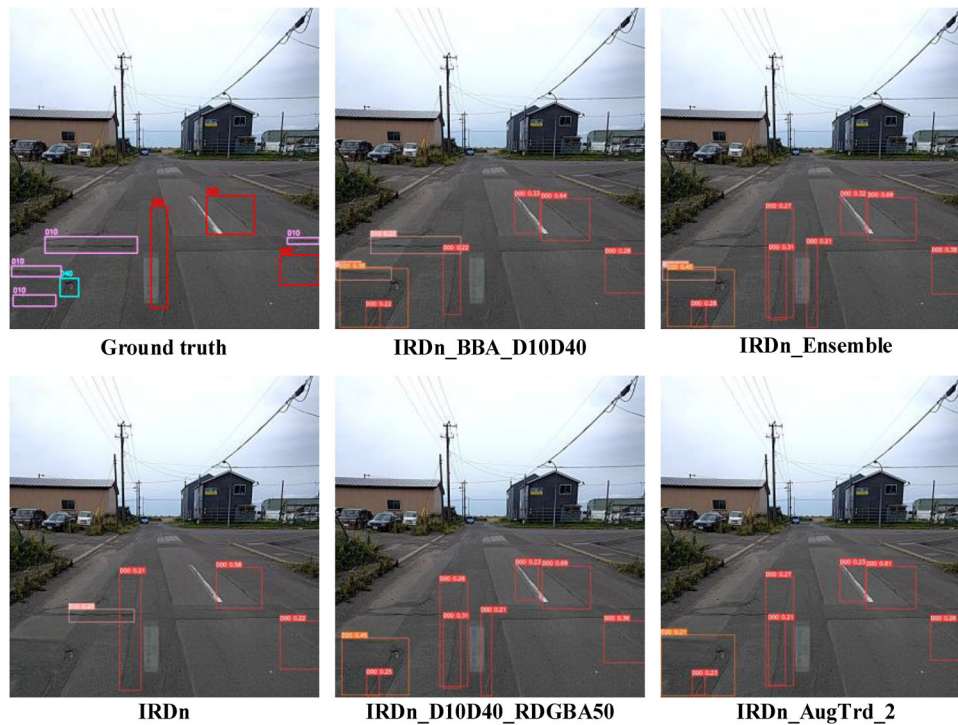


FIGURE 19 Detection samples obtained by different trained models in RDD2020.

These results indicate that the use of TTA increases the detection time of all the considered models. By applying the TTA, the “IRDn_Ensemble” model has improved the detector precision in all the road damage classes relative to the reference model. Also, the accuracy of the “IRDn_Ensemble” model is similar to that of the IRDx model, but is 35% faster than the latter model. The results of this table also show that the “IRDn_Ensemble” model has a much lower floating point operations per second (FLOPS) requirement than the IRDx model and thus can be used on devices of low processing power for the real-time detection of road damages. Some examples of the damages detected by each model have been displayed in Figure 18. The performances of the detectors in this paper on some sample images of the RDD2020 dataset have been depicted in Figure 19. For more exact results, these models have to be trained with the RDD2020 dataset.

6 | CONCLUSION

In this paper, the process of collecting road damage data was improved by presenting a technique that automatically discards the inferior and low-quality data. Next, by employing this same method, a dataset of Iran road damages was produced. This dataset contains 25000 annotated images. Then the performances of the YOLOv5-based object detectors in the detection of road surface damages were examined. Various types of data augmentation techniques were used to boost the accuracy, precision and the generalization of the detection models. The results

of the models trained via traditional augmentations show the greater generalization of these models. We subsequently presented the bounding box augmentation and generated new data by applying this approach. This method enhanced the precision of the models in the augmented classes. Because of the restrictions in collecting the data in the D10 and D40 damage classes, a “road damage GAN based augmenter” was presented. In this method, by applying the automatic road damage blender algorithm and using the real and the synthetic data generated by GANs, new images were produced. These extra images compensate, to some extent, the shortages that exist in certain road damage classes, such as D40. However, using all the synthetically produced images makes the detection models biased. Finally, by ensembling these models, a road damage detection model was created which, in the developed dataset, performs better than a reference model. This model can also perform better than a reference model in different lighting conditions, shades, and sun flare settings.

Recording road scene images by means of a mobile phone camera provides relatively low-resolution pictures for road damage detection purposes. Moreover, using the data collected by a sensor does not ensure an error-free detection at all times and under all environmental and ambient conditions. Using a combination of vibration sensor data and camera image information can be an appropriate means of detecting potential road damages. In case information is collected via “crowdsourcing”, a procedure should be devised for making the gathered data uniform and for measuring the information quality. The standardization of this process helps to establish a uniform dataset of good quality. In the future

works, we would like to improve the performance of the proposed method by implementing the ensemble learning techniques such as bagging and boosting. We also want to use different sensors to investigate the performance of road damage detectors in online detection conditions in autonomous vehicles.

AUTHOR CONTRIBUTIONS

Nima Aghayan-Mashhady: Data curation, Formal analysis, Investigation, Software, Visualization, Writing—original draft. Abdollah Amirkhani: Conceptualization, Methodology, Supervision, Validation, Writing—original draft, Writing—review & editing.

CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

The dataset and code are available at: <https://github.com/IranRoadDamageDataset/IRRDD>

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